Lending Under Imperfect Information

Final update: discrimination and predictive lending

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3 November 2026

How does direct discrimination affect bank profit, and how does gradient boosting compare with OLS?

Group A (B=0). Minority Group B (B=1). Selected penalty: +0.005B.

80,000 training. 20,000 prediction. Profits only from the prediction subset.

Single matched synthetic run. Penalty calibrated to prediction-group approvals for illustration.

Training, Prediction and Bank Profit

80,000 TRAINING applicants
Observed characteristics, credit proxy C
and historical-style noisy PN

→

20,000 PREDICTION applicants
Frozen OLS and gradient-boosting models
produce default-risk scores

The bank cannot observe true PD or every risk determinant.

PN = 0.8 true_PD + η
C is a noisy proxy for the hidden risk factor.

OLS control / OLS + 0.005B / ML
Bounded PD = clip(raw score, 0, 1)

↓

Prediction-subset
expected cash flows

←

Expected loan profit
Σ c_t (1–p)^t – L

←

Capital-constrained
common lending cutoff

All approvals and all reported profits use prediction applicants only

Working contract: T=30, theta=0.10, r=5% per period. Fixed capital $1.826087B.

Scenario 1: No Systemic Differences

Group A and minority Group B share the same underlying risk process

The bank observes Y, D and C to estimate default risk.

The direct-discrimination branch adds +0.005B after the OLS fit and before probability clipping.

The added group penalty is arbitrary relative to the group-neutral true-risk process.

Same group-neutral generating process. Finite realized group averages may differ.

Scenario 1: Three Lending Methods

Higher estimated default risk lowers lending priority

Method	Raw score	Interpretation
OLS control	f_OLS(Y, D, C)	Naïve risk estimation
OLS + penalty	f_OLS(Y, D, C) + 0.005 B	Direct discrimination: +0.5 pp for B=1
Gradient boosting	f_ML(Y, D, C)	Flexible predictive estimation

Training: 80,000. Prediction and profit: 20,000.

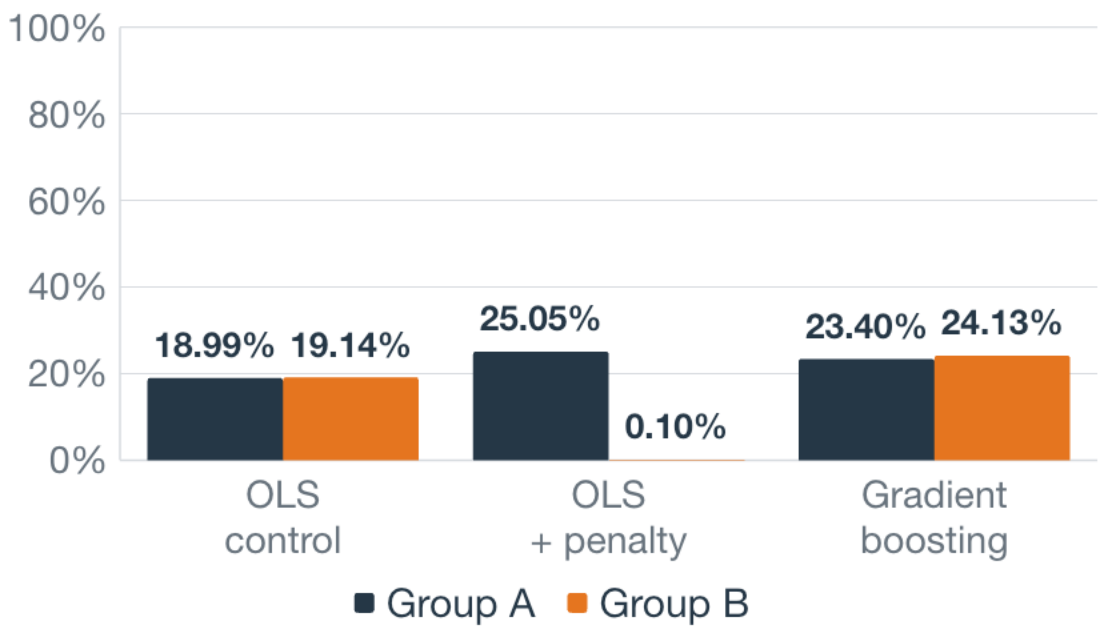
Penalty selected from prediction-group approvals for illustration. True profit is not a selection criterion.

Same permitted information within each scenario. Historical models and prediction scores remain frozen.

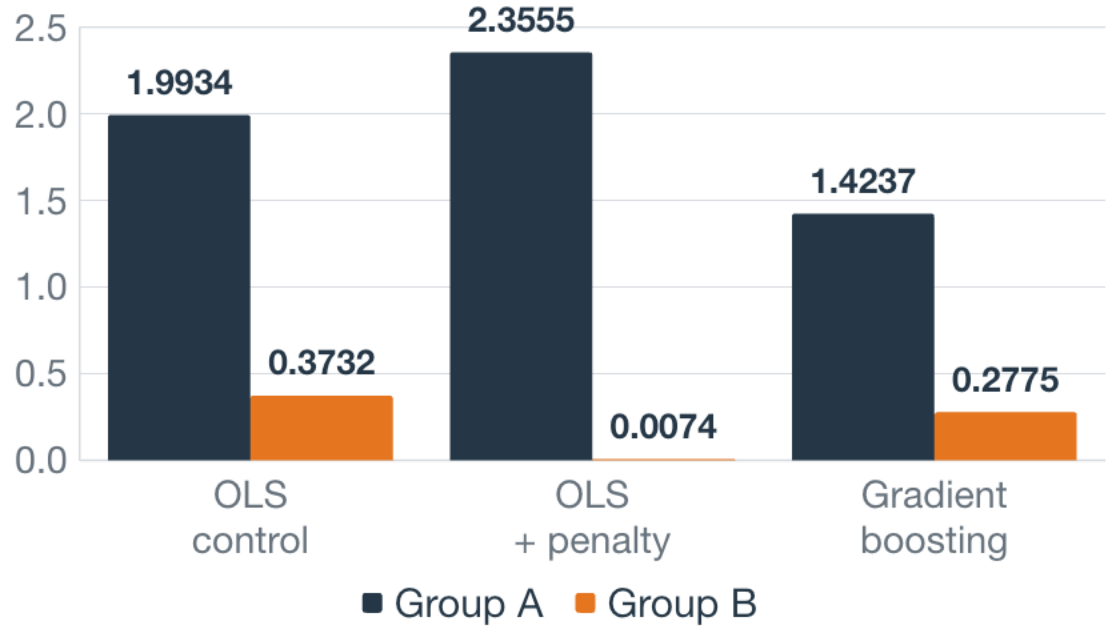
Scenario 1: Prediction-Subset Results

Profits: prediction subset only. Gradient boosting uses 72.4% of capital.

Within-group approval rate



True expected profit by group ($B)



Penalty +0.005B: A 4,251/16,970 (25.05%); B 3/3,030 (0.10%).

Penalty minus control: true expected portfolio profit –$3.733M

Source: small_penalty/group_results.csv and portfolio_results.csv. Same contracts and capital.

Scenario 2: Systemic Differences

Group B has a higher hidden risk factor from a process outside the lending decision

$\varepsilon_2 = \varepsilon_1 + B$

$C_2 = 0.4\ \varepsilon_2 + \omega$
$PN_2 = 0.8\ true_PD_2 + \text{the same } \eta$

OLS and Gradient Boosting use Y, D, C and B

Group B mean true PD: 0.14229% to 0.14476%.
The risk function is unchanged. Between-world results also change the permitted predictors.

Existing +1 SD epsilon shift for minority Group B. No new population or risk-equation change.

Scenario 2: Three Lending Methods

Higher estimated default risk lowers lending priority

Method	Raw score	Interpretation
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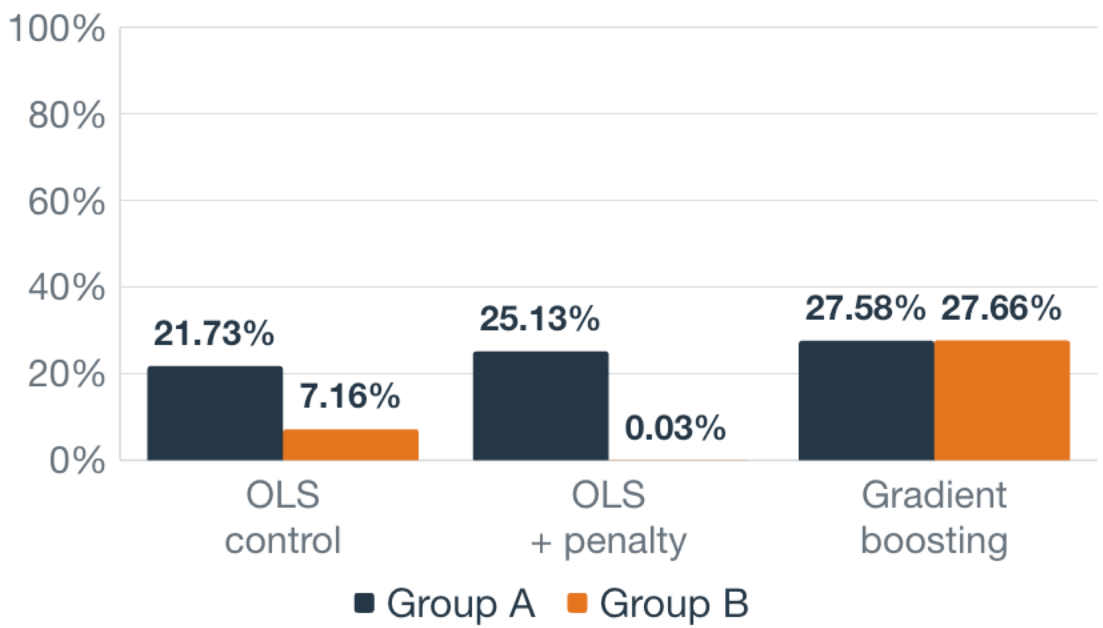
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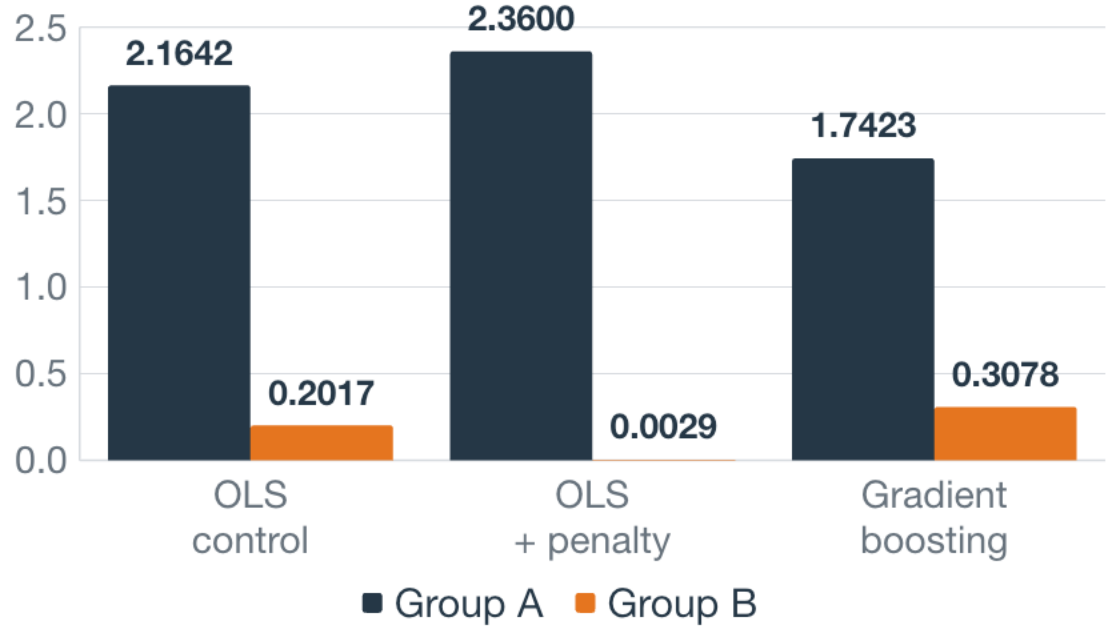
Scenario 2: Prediction-Subset Results

Profits: prediction subset only. Gradient boosting uses 87.2% of capital.

Within-group approval rate



True expected profit by group ($B)



Penalty +0.005B: A 4,265/16,970 (25.13%); B 1/3,030 (0.03%).

Penalty minus control: true expected portfolio profit –$2.983M

Source: small_penalty/group_results.csv and portfolio_results.csv. Same contracts and capital.

Moving the Tie Leaves Lending Unchanged

Replace only exact zeros with a predeclared positive floor

Quantity	Original direct policy	Positive-floor sensitivity
Shared tie score	0%	0.013924%
Applicants in the tie	3,764	3,764
Total approvals	4,990	4,990
True portfolio profit	$2.3593B	$2.3593B

0 approval decisions change

Perceived profit changes by –$7.855M. The tie remains intact.

Source: mechanism_audit/positive_floor_derivation.json and portfolio_comparison.csv. Canonical sensitivity.

Removing B's Contribution Without Refitting

raw_Z = raw_W – saved_beta_B × B

W keeps B. Z removes only its contribution. F is the full blind refit.

Approval gap, B=1 minus B=0 (pp)



Z minus W true profit

+$853,049

Z minus F true profit

+$327,762

Z and F have equal group counts but exchange one approved loan.

Source: fixed_coefficient/cd7f3ee268cfc4d0/canonical_policies.csv. One canonical realization.

No Universal Profit Gain Across 200 Draws

Share of draws by signed true-profit difference



Comparison	Mean change	Median change	MCSE of mean
Z minus W	+$12.82M	+$0.593M	$20.53M
Z minus F	–$11.80M	$0	$26.50M

Rare reject-all disagreements dominate the means

Source: fixed_coefficient/cd7f3ee268cfc4d0/noise_summary.csv. Same population and 200 paired saved noise draws.

Ranking Scores and Accounting Probabilities

Role	Original bounded policy	Raw-order sensitivity
Lending order	clip(raw, 0, 1)	raw score
Cash-flow PD	clip(raw, 0, 1)	Same clip(raw, 0, 1)
Indivisible ties	Exact bounded scores	Exact raw scores

Draw 008, aware: the zero-score block requires 110.01% of capital.

Raw ordering permits a feasible prefix within the clipped block

Every evaluation request is perceived as profitable, so capital limits lending.
Raw scores rank requests. Cash flows use the same bounded PD.

Source: raw_ordering.yaml and saved raw-ordering bank freezes. Fixed probabilities, contracts and capital.

Nine Reject-All Policies Now Fund Loans

Approvals shown as original / raw ordering

Case	W: aware	Z: B removed	F: blind refit
Canonical	3,900 / 3,900	3,802 / 3,802	3,802 / 3,802
Draw 008	0 / 5,107	0 / 4,741	4,744 / 4,744
Draw 064	0 / 3,898	3,701 / 3,701	0 / 3,701
Draw 089	4,331 / 4,331	0 / 4,163	4,161 / 4,161
Draw 092	0 / 4,222	0 / 3,865	3,861 / 3,861
Draw 154	0 / 4,812	3,454 / 3,454	0 / 3,454

Risk estimates and per-request profits stay unchanged

The aware approval gap in draw 154 remains +88.25 pp.

Source: raw_ordering/d0421da2ea691f2e/policies.csv. Six selected diagnostics, not a random uncertainty sample.

Decisions Before the Final Research Report

Lending order	May raw scores rank requests while bounded PD determines profits?
Contract interpretation	Is 5% per period with the 30-period growing schedule intended?
Measurement noise	Keep SD 0.1 as the working case, or predeclare a separate noise study?
Paper scope	Should the final paper add an ML benchmark for this model?

Next deliverable: a final analysis protocol and paper outline